Lecture: Sets and Functions

Mathematical Economics — Core Tools

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Contents

1 Sets

Definition 1 (Set). *A* set *is a collection of distinct objects, called elements. If x is an element of a set A, we write $x \in A$; otherwise $x \notin A$.*

1.1 Describing Sets

- **Roster notation:** $A = \{1, 2, 3\}$.
- **Set-builder notation:** $B = \{x \in \mathbb{Z} : x \text{ is even}\} = 2\mathbb{Z}$.
- **Intervals in $\mathbb{R}$:** (a, b), $[a, b]$, $(a, b]$, $[a, b)$.

Example 1. *Let $S = \{(i, j) \in \mathbb{Z}^2 : i + j = 4\}$. Then $S = \{(0, 4), (1, 3), (2, 2), (3, 1), (4, 0)\}$.*

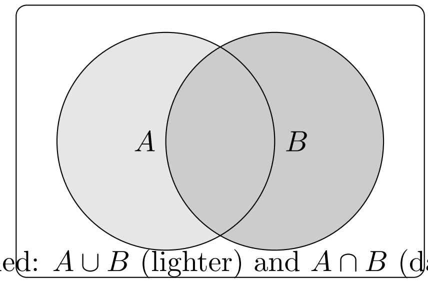
1.2 Subsets and Equality

Definition 2 (Subset)**.** *$A \subseteq B$ if every element of A is in B. If $A \subseteq B$ and $B \subseteq A$, then $A = B$.*

Proposition 1 (Basic Laws)**.** *For any sets A, B, C in a universe U:*

1. *$A \cap B = B \cap A$ (commutativity), $A \cup B = B \cup A$.*
2. *$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (distributivity), similarly for $\cup$ over $\cap$.*
3. *$(A^c)^c = A$, $A \cup A^c = U$, $A \cap A^c = \varnothing$ (complements).*

1.3 Operations with Venn Diagrams



Shaded: $A \cup B$ (lighter) and $A \cap B$ (darker)

1.4 Power Set and Cartesian Product

Definition 3 (Power set)**.** *The* power set *of A is $\mathcal{P}(A) = \{B : B \subseteq A\}$.*

Example 2. *If $A = \{a, b\}$ then $\mathcal{P}(A) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}$.*

Definition 4 (Cartesian product)**.** *$A \times B = \{(a, b) : a \in A,\ b \in B\}$.*

2 Relations and Functions

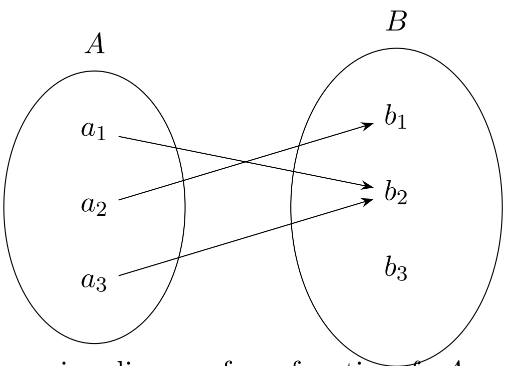
2.1 Relations

A relation R from A to B is a subset of $A \times B$. We write $a\,R\,b$ if $(a, b) \in R$.

Example 3. *On $\mathbb{Z}$, the relation $\leq$ is a subset of $\mathbb{Z} \times \mathbb{Z}$: $\leq = \{(m, n) : m \leq n\}$.*

2.2 Functions

Definition 5 (Function)**.** *A function f from A to B, written $f\colon A \to B$, assigns to each $a \in A$ a unique element $f(a) \in B$. Here A is the* domain *and B a* codomain*. The* image *(range) is $\operatorname{im}(f) = \{f(a) : a \in A\} \subseteq B$.*



A mapping diagram for a function $f\colon A \to B$

2.3 Images and Preimages

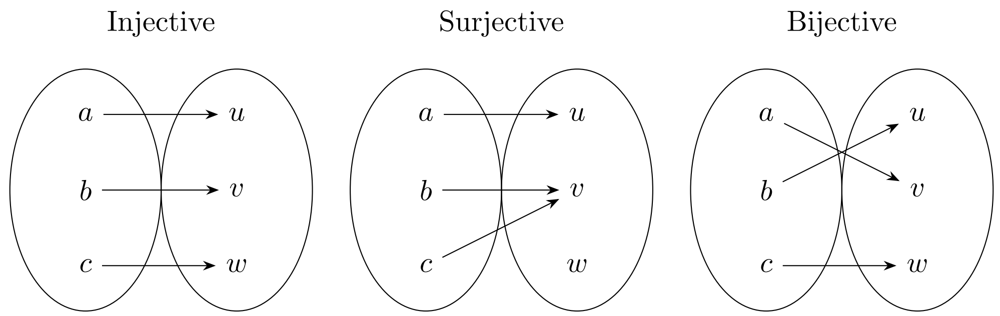
For $S \subseteq A$, define $f[S] = \{f(a) : a \in S\} \subseteq B$. For $T \subseteq B$, the preimage is $f^{-1}[T] = \{a \in A : f(a) \in T\}$.

Example 4. *Let $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Then $f[(-2, 1)] = [0, 4)$ and $f^{-1}[(1, 4]] = [-2, -1) \cup (1, 2]$.*

2.4 Injective, Surjective, Bijective

Definition 6. *A function $f\colon A \to B$ is*

- injective *(one-to-one) if $f(a) = f(a') \Rightarrow a = a'$;*
- surjective *(onto) if* $\mathrm{im}(f) = B$*;*
- bijective *if it is both injective and surjective.*



2.5 Inverse and Composition

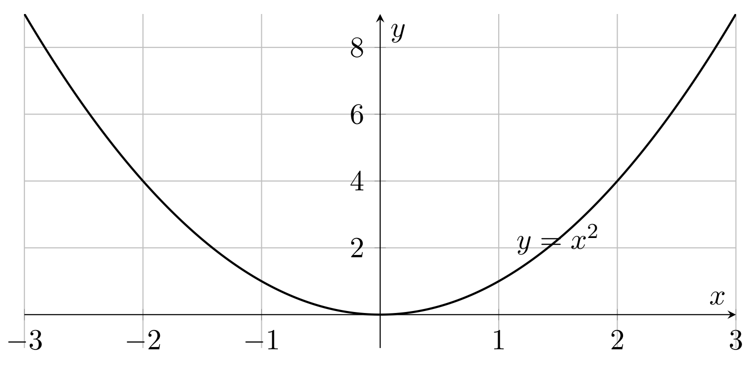
Definition 7 (Inverse)**.** *If $f\colon A \to B$ is bijective, the* inverse $f^{-1}\colon B \to A$ *satisfies $f^{-1}(f(a)) = a$ and $f(f^{-1}(b)) = b$.*

Definition 8 (Composition)**.** *If $f\colon A \to B$ and $g\colon B \to C$, then $(g \circ f)\colon A \to C$ is $(g \circ f)(a) = g(f(a))$.*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend left, "g \circ f"] & B \arrow[r, "g"] & C \end{tikzcd}$$

2.6 Graphs of Real-Valued Functions

A function $f\colon \mathbb{R} \to \mathbb{R}$ can be represented by its graph $\{(x, f(x)) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$.



3 Worked Examples

Example 5 (Counting with Power Sets). *How many subsets does an n-element set have?* Answer: $|\mathcal{P}(A)| = 2^n$*, since each element can be either included or excluded.*

Example 6 (Injectivity and Surjectivity). *Consider $f\colon \mathbb{Z} \to \mathbb{Z}$, $f(n) = 2n + 1$. Then f is injective (since $2n + 1 = 2m + 1 \Rightarrow n = m$) but not surjective onto $\mathbb{Z}$ (even integers are not in the image).*

Example 7 (Images and Preimages). *Let $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$. For $S = [-1, 2]$, we have $f[S] = [-1, 8]$. For $T = [1, 27]$, $f^{-1}[T] = [1, 3]$.*

4 Useful Proof Patterns

4.1 Proving Set Equality

To prove $A = B$, show $A \subseteq B$ and $B \subseteq A$.

4.2 Proving Injectivity/Surjectivity

- **Injective:** Assume $f(a) = f(a')$ and deduce $a = a'$.
- **Surjective:** For each $b \in B$, construct an $a \in A$ with $f(a) = b$.

5 Mini Exercises (with brief solutions)

E1: Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. List $A \times B$.

Solution: $\{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$.

E2: Prove or disprove: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution sketch: Use element-chasing in both directions.

E3: Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2 + 1$. Is f injective? Surjective?

Solution: Not injective (since $f(1) = f(-1)$). Not surjective onto $\mathbb{R}$ (range is $[1, \infty)$).

E4: If $f\colon A \to B$ and $g\colon B \to C$ are injective, prove $g \circ f$ is injective.

Solution: $(g \circ f)(a) = (g \circ f)(a') \Rightarrow g(f(a)) = g(f(a')) \Rightarrow f(a) = f(a') \Rightarrow a = a'$.

6 Connections to Mathematical Economics

In general equilibrium and trade theory, sets formalize choice sets and feasible allocations; functions encode preferences ($u\colon X \to \mathbb{R}$), technologies ($F\colon \mathbb{R}^n_+ \to \mathbb{R}_+$), and excess demand ($z\colon \mathbb{R}^n_{++} \to \mathbb{R}$). Injectivity relates to identification, surjectivity to existence, and bijections to change-of-variable arguments in welfare comparisons.

End of lecture notes.